

Experiment: Jar Test for Optimum Coagulant Dose

Coagulation–Flocculation of Tannery Process Liquors using Ferric Chloride and Potash Alum

Reference standard: ASTM D2035-19, Standard Practice for Coagulation-Flocculation Jar Test of Water

Aim

To determine, by the jar test, the optimum coagulant and dose that best clarify a tannery process liquor — comparing ferric chloride with potash alum — and to judge the result from the removal of turbidity/colour of the settled supernatant.

Definitions & Key Terms

Understand these terms before starting; they explain why the test works.

Colloid — an extremely small particle (about 1 nm to 1 μm) that stays suspended in water and does not settle under gravity, because it is so light and carries a surface electrical charge. Colloids cause the turbidity and much of the colour of effluent.

Turbidity — the cloudiness of water caused by suspended and colloidal particles scattering light. The clearer the water, the lower the turbidity. Here it is followed as the absorbance measured on the UV-Vis spectrophotometer.

Coagulation — the chemical destabilisation of colloids by adding a coagulant that neutralises their surface charge, so the repulsion disappears and the particles can approach one another. It requires fast, intense mixing (rapid mix).

Coagulant — a chemical — usually a trivalent metal salt such as ferric chloride or alum — that supplies highly charged metal ions (Fe^{3+} , Al^{3+}) to neutralise the colloid charge and that forms a metal-hydroxide precipitate.

Flocculation — the slow, gentle mixing that follows coagulation, allowing the destabilised particles to collide and grow into larger, visible aggregates. Too vigorous a stir would tear these apart.

Floc — the larger, settleable clump formed when destabilised colloids join together with the metal-hydroxide precipitate. Big, dense flocs settle quickly.

Coagulant aid (flocculant / polyelectrolyte) — a long-chain polymer that bridges particles to make bigger, stronger flocs. It is not used in this experiment but you should know the term.

Sweep-floc coagulation — a second mechanism in which the growing metal-hydroxide precipitate physically enmeshes and drags down colloids as it settles — like a net sweeping through the water.

Jar test — a bench-scale imitation of the full-scale coagulation → flocculation → sedimentation process, run in beakers (“jars”) to find the best coagulant and dose before applying them in the plant (ASTM D2035).

Optimum dose — the lowest coagulant dose that gives the best clarification. Below it, coagulation is incomplete; above it, the colloids can pick up the opposite charge and re-disperse (overdosing / restabilisation).

Supernatant — the clear liquid left above the settled sludge, which is sampled for the turbidity/colour measurement.

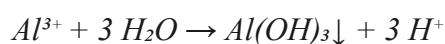
Principle / Theory

Colloidal particles in effluent carry like electrical charges and repel each other, so they never collide and never settle — the water stays cloudy and coloured. A coagulant supplies highly charged metal ions that cancel this repulsion (charge neutralisation) and, on hydrolysis, form a gelatinous metal-hydroxide precipitate that both traps colloids (sweep-floc) and grows into settleable flocs. The two steps are always paired: a fast mix to disperse the coagulant and neutralise charge (coagulation), then a slow mix to grow the flocs (flocculation), followed by gravity settling.

In plain terms: coagulation is like adding lemon to milk to make it curdle; flocculation is the gentle stir that gathers the curds into lumps you can remove.

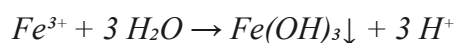
The two coagulants compared

Potash alum — potassium aluminium sulphate, $KAl(SO_4)_2 \cdot 12H_2O$. In water the Al^{3+} ion hydrolyses to form aluminium hydroxide floc and releases acid:



It works best in a narrow pH band of about 6–7.5 and lowers the pH as it acts.

Ferric chloride — $FeCl_3$. The Fe^{3+} ion hydrolyses to ferric hydroxide floc, also releasing acid:



It works over a wider pH range (about 5–9), gives denser, faster-settling flocs, and is especially good at removing colour — but it can leave a residual colour of its own. Both coagulants consume alkalinity and drop the pH, so pH must be watched.

Significance

Coagulation–flocculation is the chemical heart of primary treatment. It removes colloids, part of the COD, colour and residual chromium before sedimentation. Each tannery liquor behaves differently (very different pH and pollutants), so the best coagulant and dose must be found by test, not assumed.

Group Allocation — Four Groups

Each group tests ONE tannery process liquor and, on that liquor, compares BOTH coagulants (ferric chloride and potash alum) across a dose series. Pooling the four groups shows which coagulant and dose suit each stream.

Group	Assigned process liquor (both coagulants)	Members
Group 1	Soaking liquor — salt, dirt, soluble protein, high solids	_____
Group 2	Liming liquor — high pH (11–12), sulphide, lime, protein	_____
Group 3	Deliming liquor — ammonium salts, lower pH, soluble protein	_____
Group 4	Chrome tanning liquor — chromium, low pH (2.5–4), sulphate	_____

Apparatus & Instruments

- **Magnetic stirrer with stir bars** — for the rapid and slow mixing (see note below)
- **UV-Vis spectrophotometer** — to measure absorbance of the supernatant as turbidity / colour
- **Multi-parameter meter** — to measure pH (and conductivity) of the liquor and supernatant
- **Beakers (500 mL), pipettes, graduated cylinders, filter papers/syringe filters** — for dosing, sampling and clarifying the supernatant

Note on the magnetic stirrer: a standard jar test uses a six-paddle gang stirrer that runs all beakers at once. With a single magnetic stirrer you must run one beaker at a time in sequence. Use the highest stable speed for the 1-minute rapid mix and the lowest stable speed for the 20-minute slow mix. Keep the same speeds and stir-bar size for every beaker so the results are comparable (ASTM D2035 requires identical conditions).

Reagents

- **Ferric chloride stock, 10 g/L** — 1 mL = 10 mg FeCl₃
- **Potash alum stock, 10 g/L** — 1 mL = 10 mg alum
- **Dilute H₂SO₄ and lime / NaOH** — to adjust pH into the coagulant's working range if needed
- **Distilled water** — for dilutions and the spectrophotometer blank

Because tannery liquors are strong and dark, the supernatant will usually need dilution to read on the spectrophotometer. Use the same dilution for the raw liquor and for every treated sample so the readings can be compared.

Preparing the stock and working out the dose

You add the coagulant as a solution, never as raw powder — powder cannot be dosed accurately or dispersed fast enough. Make a 10 g/L stock of each coagulant once (weigh 10 g of powder, dissolve, and make up to 1 litre). Then 1 mL of stock contains 10 mg of coagulant.

$$\text{Volume of stock to add (mL)} = \text{dose (mg/L)} \times \text{beaker volume (L)} \div 10$$

For the 500 mL (0.5 L) beaker used here, the three doses work out as:

Target dose (mg/L)	Coagulant needed in 500 mL (mg)	Volume of 10 g/L stock to add (mL)
200	100	10
400	200	20
600	300	30

Use the same three volumes for both ferric chloride and potash alum. (If you use a 250 mL beaker instead, halve every volume: 5, 10 and 15 mL.)

How many beakers? Only SIX per group — three doses for each of the two coagulants. The magnetic stirrer mixes one beaker at a time, but settling needs no stirrer: once a beaker finishes its slow mix, set it aside to settle for 30 minutes while you mix the next one. Staggering this way keeps the single stirrer from becoming a bottleneck.

Procedure

A. Characterise the raw liquor

1. Measure and record the raw liquor's pH and conductivity on the multi-parameter meter.
2. On the UV-Vis, scan a diluted sample of the raw supernatant to find the wavelength of maximum absorbance ($\lambda_{\max}$). Measure and record the raw absorbance, A_0 , at this wavelength. *Use this same wavelength for all readings.*

B. Run the dose series (repeat for EACH coagulant)

3. Measure 500 mL of the liquor into a beaker with a stir bar.
4. Add the volume of coagulant stock for the beaker's target dose, taken from the dosing table (10, 20 or 30 mL for 200, 400 or 600 mg/L). If the liquor's pH is far from the coagulant's working range, adjust it first and note the value.
5. Rapid mix at the highest stable speed for 1 minute. This disperses the coagulant and neutralises the colloid charge.
6. Slow mix at the lowest stable speed for 20 minutes. This lets the flocs grow; a gentle stir avoids breaking them.
7. Stop the stirrer and let the beaker settle undisturbed for 30 minutes.
8. Draw supernatant from a fixed depth, filter if needed, dilute as in step A, and measure its absorbance (A) and pH.
9. Repeat for the three doses (200, 400, 600 mg/L), then repeat the three with the second coagulant — six beakers in all.

Keep beaker size, volume, stir-bar, speeds, times and settling time identical throughout — only the coagulant type and dose change.

Observations

A. Your own group's data

Raw liquor: pH = _____ conductivity = _____ $\lambda_{\max}$ = _____ nm A_0 = _____

Ferric chloride series

Dose (mg/L)	Final pH	Supernatant absorbance A	Removal (%)
0 (raw)			—
200			
400			
600			

Potash alum series

Dose (mg/L)	Final pH	Supernatant absorbance A	Removal (%)
0 (raw)			—
200			
400			
600			

B. Pooled class results (copy every group's best result)

Process liquor	Best coagulant	Optimum dose (mg/L)	Max removal (%)
Soaking			
Liming			
Deliming			
Chrome tanning			

Calculations, Graph & Optimum Dose

$$\text{Removal (\%)} = [(A_0 - A) / A_0] \times 100$$

where A_0 = absorbance of the raw supernatant and A = absorbance of the treated supernatant (same wavelength and dilution).

For each coagulant, plot removal (%) against dose. The optimum dose is the lowest dose that reaches the maximum removal — the point where the curve levels off. Compare the two curves: the coagulant giving higher removal at a lower dose (and acceptable final pH) is the better choice for that liquor.

Precautions

- Keep all mixing speeds, times, volumes and settling time identical between beakers (ASTM D2035).
- Dose coagulant immediately before the rapid mix; start timing together.
- Use the same wavelength and same dilution for A_0 and every A .
- Sample all supernatants from the same depth and time.
- Watch the pH — both coagulants lower it; adjust liming (high pH) and chrome (low pH) liquors into the working range and record it.
- Handle chrome liquor with care (chromium) and liming liquor in a fume hood (sulphide / H_2S).

Result & Interpretation

Report the optimum coagulant and dose for your liquor, the maximum removal achieved, and the final pH. In the discussion, explain why one coagulant beat the other for your particular stream — for example, ferric chloride often wins on the coloured, high-pH liming liquor and on chrome-bearing liquor, while alum can suffice on the lighter soaking liquor — and relate the best pH to each coagulant's working range.